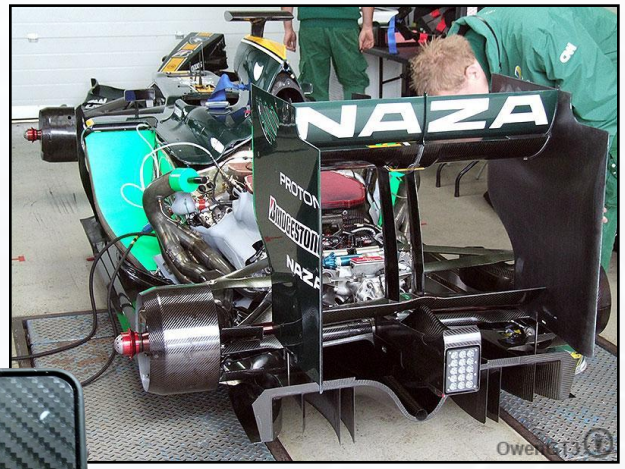
Mr Dimelow

3.1 Engineering materials

3.1.1.3 Composites

Subject content	Additional information	Mathematical understanding
• Fibre reinforced polymers (FRP): • carbon-fibre reinforced polymer • glass reinforced plastic (GRP). • Plywood. • Medium Density Fibre board (MDF). • Oriented Strand Board (OSB). • Structural concrete.	Students will not be expected to have practical experience of working with all of these composites but exam questions could refer to any of the properties listed. Students should also be able to demonstrate knowledge and understanding of how the mechanical properties of these materials can change through the: • direction/alignment of reinforcement • matrix in which the reinforcement is placed • amount of reinforcement used • size and shape of reinforcement.	

Composite Materials









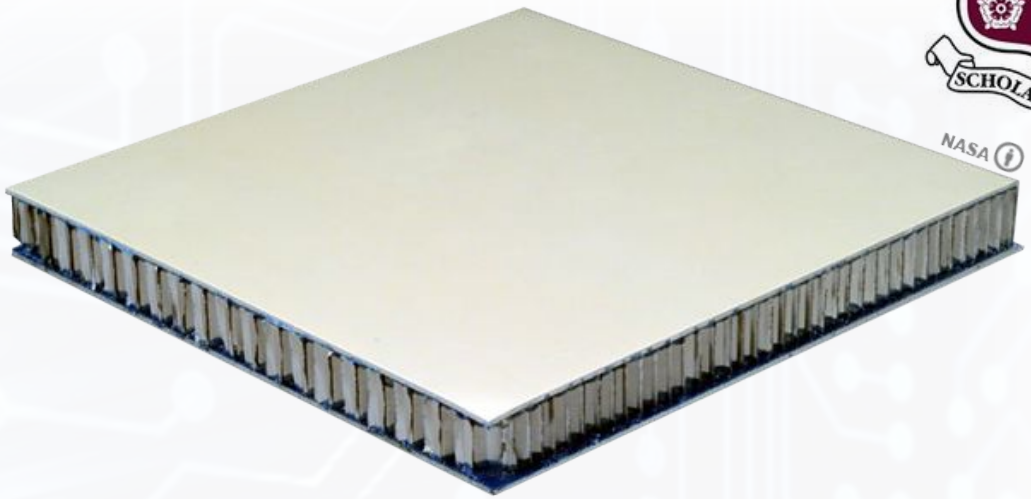
In this presentation you will:

- explore the benefits, types and examples of composite materials

Next >

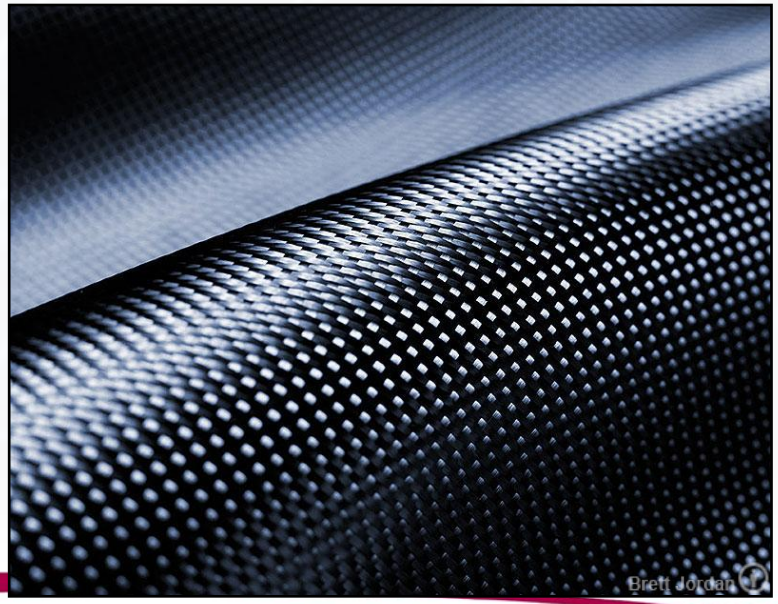
Introduction

Composite materials are made by combining two or more materials.



The resulting composite material will have a better range of properties than its constituents.

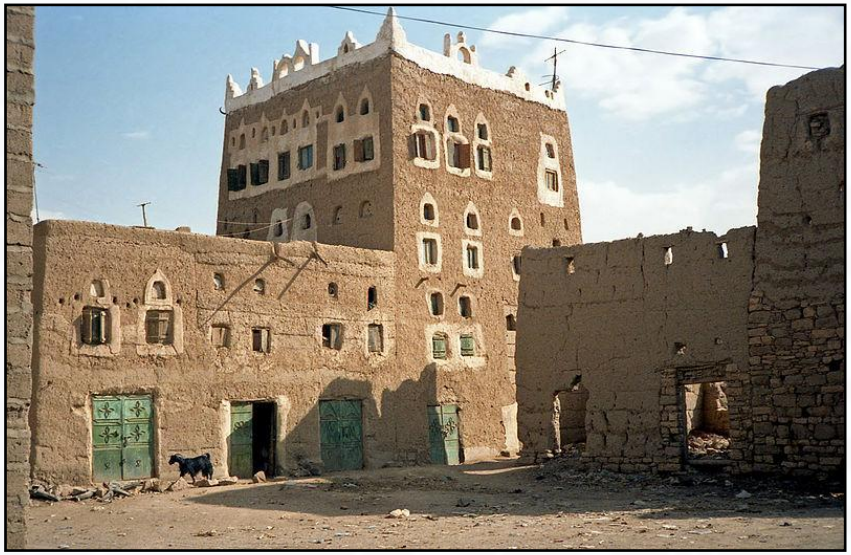
Composite materials have many modern applications.



Next >

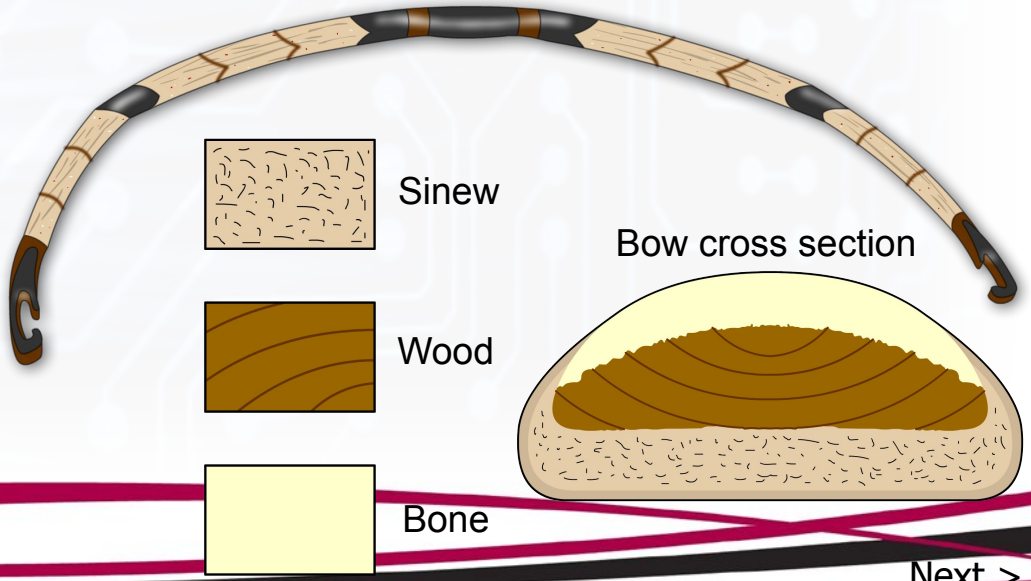
Composite Materials

We usually think of composite materials as hi-tech modern materials, but they have been in use for thousands of years.



Mud is still reinforced with straw and made into composite bricks.

Ancient steppe nomads developed composite bows made from layers of wood, horn and sinew.



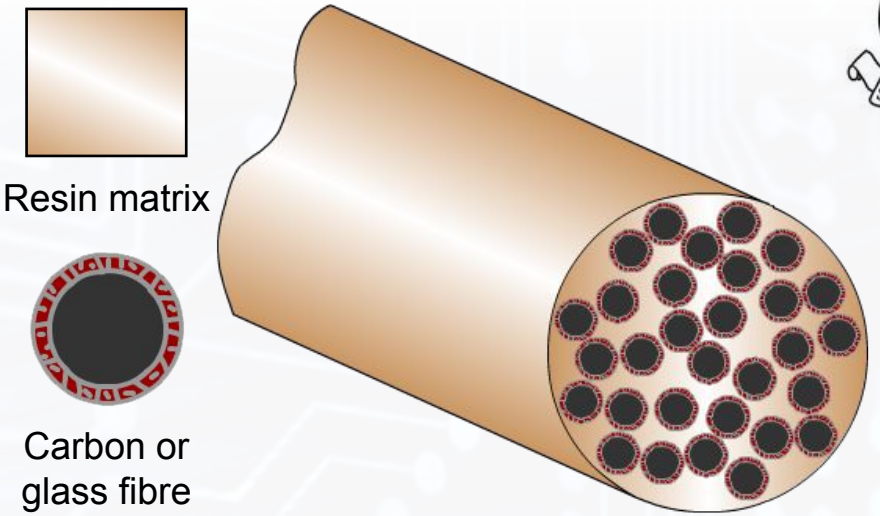
Next >

Composite Structure

Most composites consist of a bulk material, called the matrix, and a reinforcing material.

The reinforcement increases the stiffness and strength of the matrix.

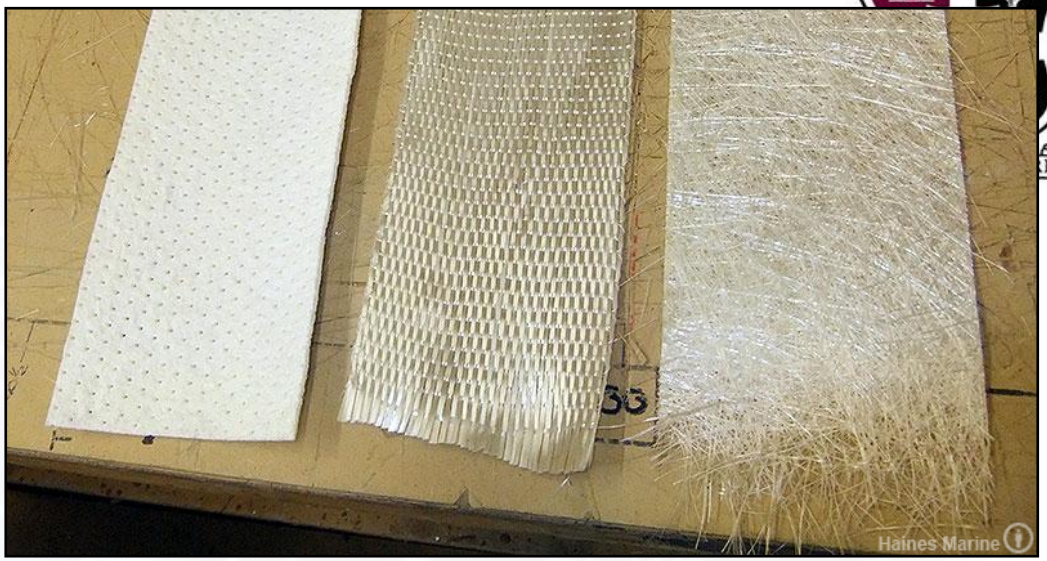
The reinforcement can also be used to impart other properties, such as hardness or corrosion resistance.





The Matrix

The matrix is the binder material that supports, separates and protects the reinforcing material.



It is continuous and allows the load to be transferred to and from the reinforcing material.

The matrix material usually has a lower density, strength and stiffness than the reinforcing material.



Next >

Types of Composite

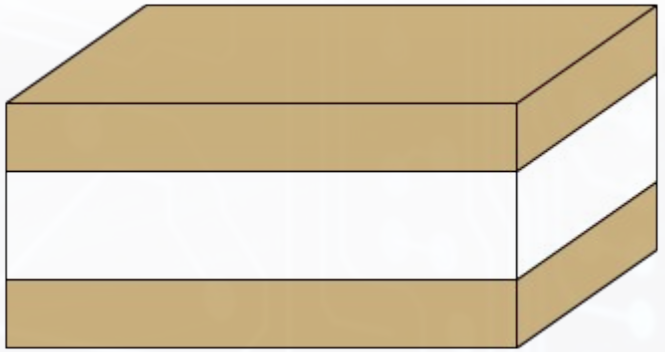
Composite materials can be grouped according to the matrix material used; polymer, metal or ceramic.

Composite materials may also be classified based on the type of reinforcing material or the structure.

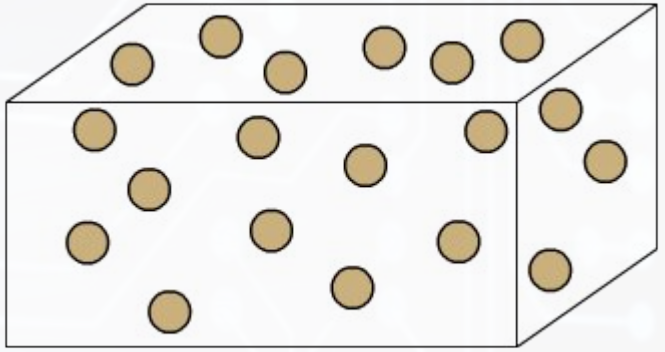
These are the main types of composite material:

- ***Laminar***
- ***Particulate***
- ***Fibre reinforced***

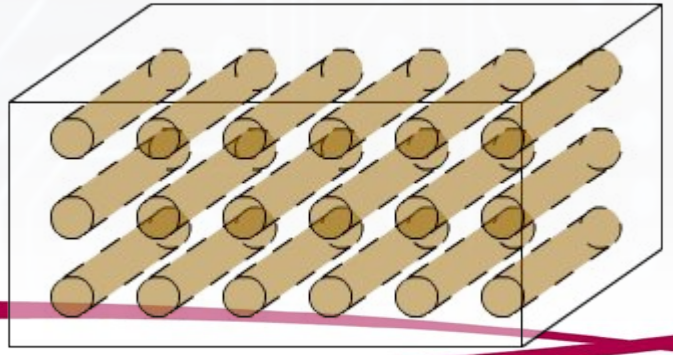
Laminar



Particulate



Fibre

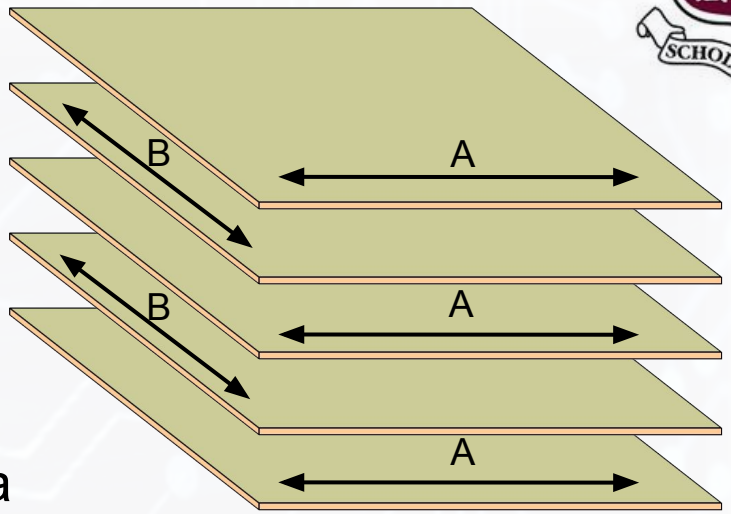


Next >

Laminar Composites

Plywood

Laminar composites are constructed from thin sheets of material bonded by the matrix material.



Plywood is a ***laminar composite*** made from thin layers of wood.

In each layer the grain is orientated at a 90° angle to the previous layer, producing a material that is equally strong in all directions.



Properties: *Smooth, even surface with good strength; often available in veneered form.*

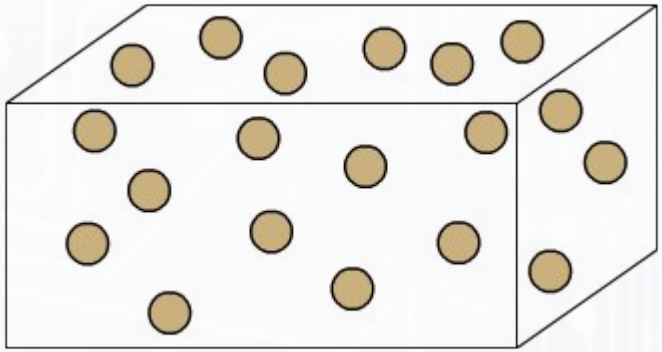
Uses: *Furniture making, structural panelling; exterior grades used for boat building*



Next >

Particulate Composites

A particulate composite consists of particles of hard brittle material suspended in a soft, ductile matrix.



These composites are designed to produce combinations of properties rather than to improve strength.



Abrasive tools are made from a particulate composite consisting of hard ceramic particles bonded in a tough polymer resin.

Next >

Particulate Composites – Cermets

Cermets are particulate composites that combine metal and ceramic materials.

Cermets are used to produce long lasting, hard wearing tools.

The ceramic component provides hardness and high temperature resistant properties.

The metal part provides flexibility, toughness and heat conductivity.





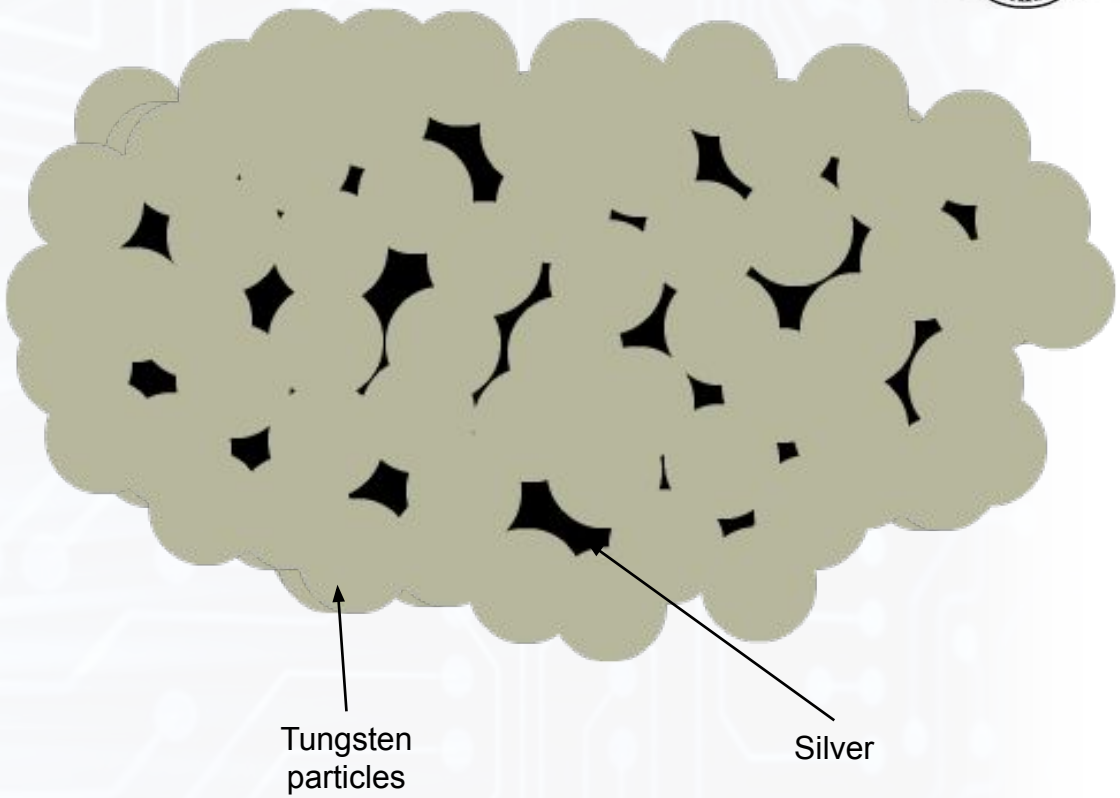
Next >

Particulate Composites – Electrical Contacts

The contacts of electrical switches need to have a combination of high wear resistance and electrical conductivity.

Tungsten powder is pressed and sintered into the required shape.

The voids in the tungsten material are then filled with silver, forming a hard wearing, electrically conductive composite.



Next >

Medium Density Fibre Board (MDF)

Medium-density fibreboard (mdf) is an engineered wood product made by breaking down hardwood or softwood residuals *(bits left over after the trees have been processed into timber products)* into wood fibres, before combining it with an ***adhesive matrix***, and forming it into panels by applying high temperature and pressure. MDF is generally denser than plywood.





Properties: *Smooth, even surface with uniform properties. Easily machined and painted; often available in veneered form. Lower cost than plywood.*



Uses: *Furniture and internal panelling.*

Next >

Oriented Strand Board (OSB)

Oriented Strand Board (OSB) is a type of engineered wood similar to particle board, formed by adding adhesives and then compressing layers of wood strands in specific orientations.



Properties: *Similar properties to plywood, but more uniform and lower cost.*

Uses: *Load bearing applications in construction, such as sheathing for walls and roof decking.*



Next >

Structural Concrete

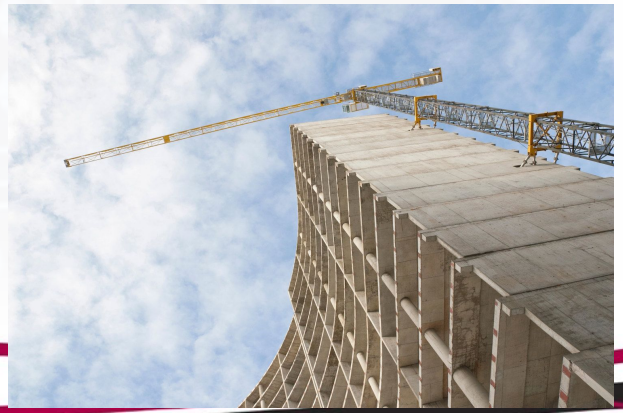
A special type of concrete that is capable of carrying a structural load or forming an integral part of a structure. The most commonly used composite material.

Properties: Excellent in compression ***concrete*** *is reinforced with steel bars to increase its tensile strength.*

Uses: *Bridges, high rise buildings*







Next >

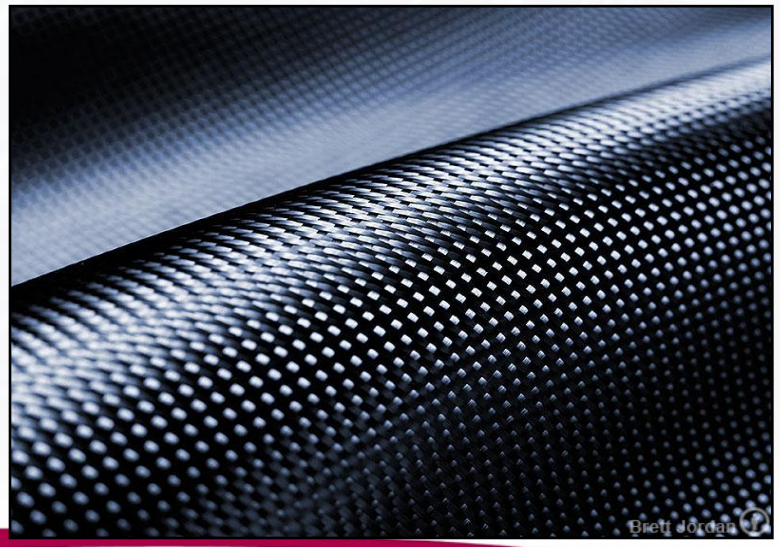
Fibre Reinforced Composites

Fibre reinforced composites (FRC) consist of long or short fibres suspended in a matrix material.

Fibre reinforced composites have improved strength, stiffness, fatigue resistance and strength to weight ratio.

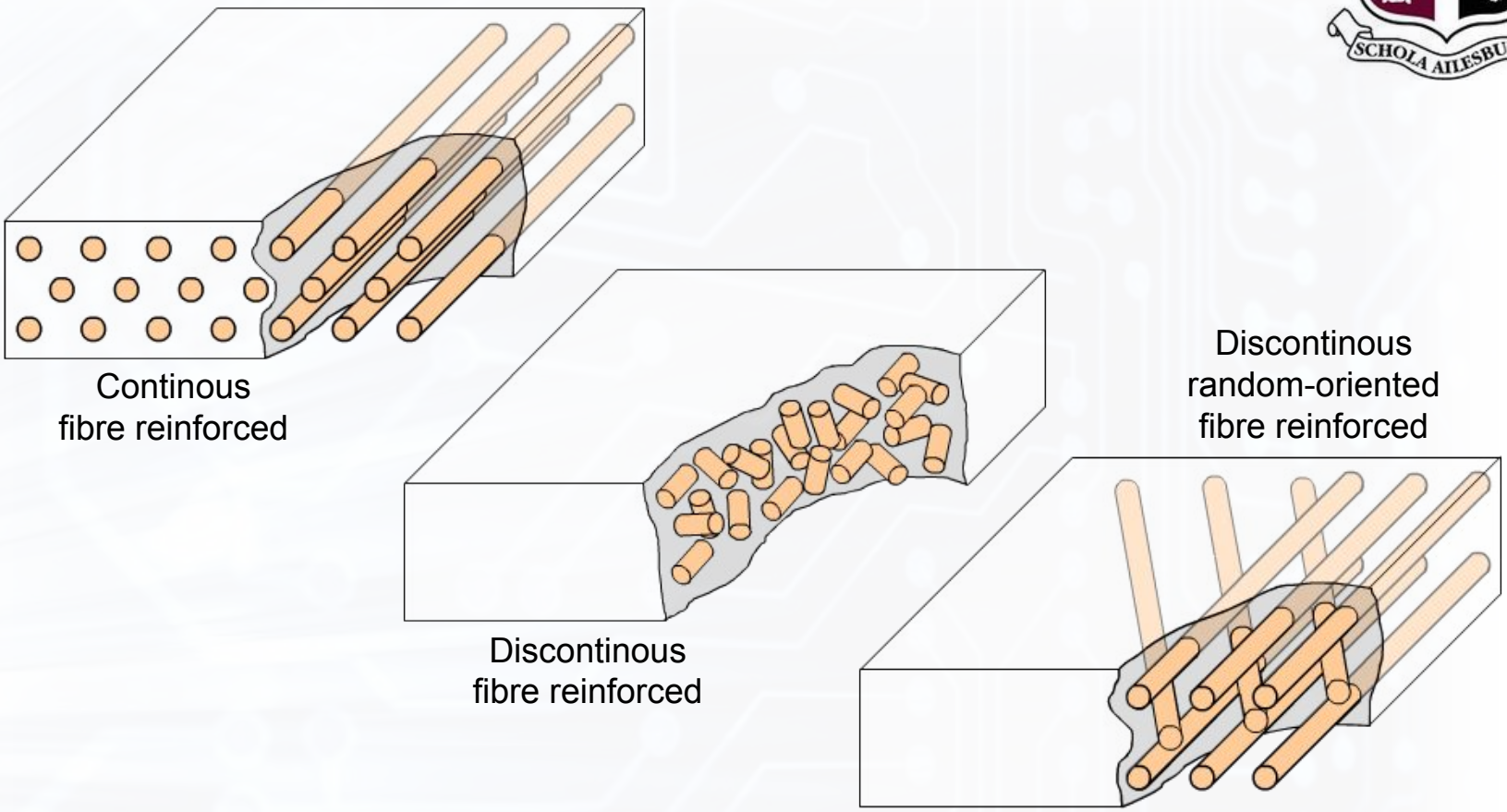
Fibres can be glass, polymer, carbon or metal.





Next >

Fibre Orientation

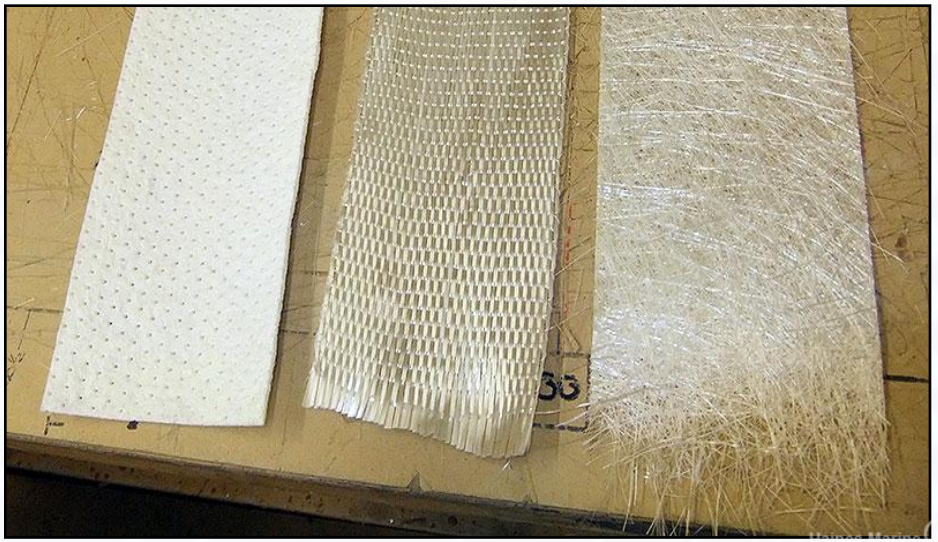


In a composite material, the fibres can be arranged in a number of ways.

Next >

Glass Reinforced Plastic

Glass reinforced plastic (GRP) consists of glass fibres in a polymer matrix, such as polyester.



The glass fibres improve the stiffness and strength of the lightweight polymer.

Glass reinforced plastic has many applications, including storage tanks, boat building and wind turbine blades.



Next >

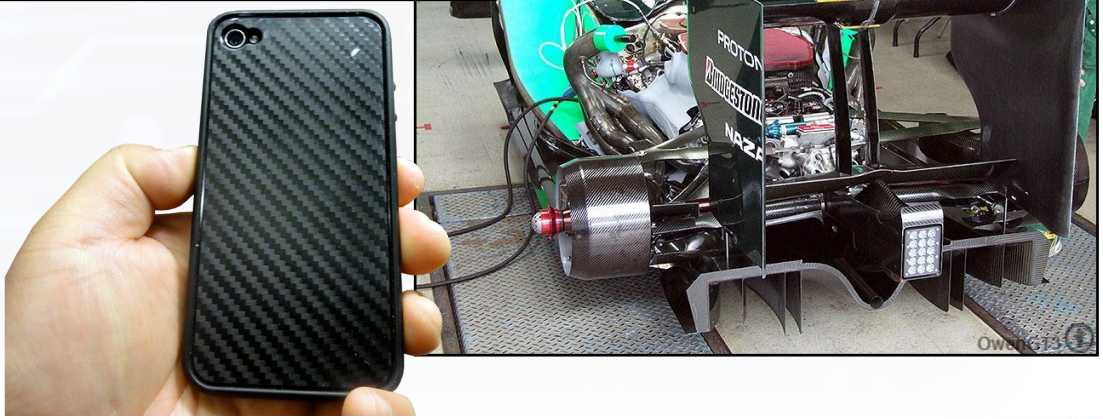
Carbon Fibre Reinforced Plastic

Carbon fibre reinforced plastic is a very strong, lightweight, composite material.



The carbon fibres have high stiffness, high tensile strength, low weight, and high temperature and chemical resistance.





Carbon fibre reinforced plastic is used in sports equipment, aerospace and high performance vehicles.

https://bikerumor.com/2019/08/26/lem-motiv-air-unveils-ultra-thin-carbon-fiber-skeleton-for-new-road-gravel-xc-bike-helmet/

Next >

Summary

In this presentation you have seen:

- how composite materials have improved properties
- examples of composite materials and their applications

End